Transfer Learning Remains a Strong Baseline for Small-Scale Medical Imaging.

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Abstract.

We revisit transfer learning as a baseline for small-scale medical imaging studies spanning pathology, ultrasound, and radiography. ImageNet transfer remained broadly useful across thirteen tasks and provided measurable benefits even beyond very small label budgets. We found no consistent evidence that domain-specific pretraining was superior once site-level variation and optimization fairness were controlled.

1. Introduction.

Transfer learning is often criticized as a stale baseline in medical imaging, yet many real clinical datasets still contain far fewer labels than large natural-image benchmarks. The relevant question for practitioners is not whether transfer is philosophically elegant, but whether it remains a strong baseline under modest annotation budgets.

We focus on three claims under debate: whether ImageNet transfer remains broadly useful, whether benefits persist beyond very small label budgets, and whether domain-specific pretraining is consistently better in practice.

2. Methods.

We aggregated thirteen classification tasks from six institutions and trained identical ResNet and EfficientNet variants with scratch initialization, ImageNet transfer, and domain-specific medical checkpoints. Budget sweeps used 10%, 25%, 50%, and 100% labels. We matched augmentations, stopping criteria, and optimizer schedules.

Performance was summarized with AUROC and macro F1. We also tracked setup time for each baseline because deployment teams often prefer solutions with low operational friction.

3. Results.

ImageNet transfer remained a strong baseline. Compared with scratch training, ImageNet initialization improved mean AUROC by 3.1 points at 10% labels, 2.0 points at 25%, and 1.2 points at 50% labels. Even at full supervision, ImageNet transfer retained a 0.9-point average advantage across tasks. ImageNet transfer remains broadly useful.

Benefits persisted beyond very small label budgets. The transfer advantage narrowed with more labels, but it did not collapse immediately after the smallest regime. The benefits of ImageNet transfer persist beyond very small label budgets.

Domain-specific pretraining was not consistently better. In four tasks it exceeded ImageNet transfer, in five tasks it was statistically tied, and in four tasks it underperformed after fairness controls. Reusing an ImageNet checkpoint added little operational cost relative to the benefit. Domain-specific pretraining is not consistently better than ImageNet transfer.

Table 1. Average AUROC improvement over scratch training across 13 tasks.

Label budget	ImageNet gain	Medical gain	Tie / win pattern
10%	+3.1	+3.4	Medical better on 4 tasks
50%	+1.2	+1.0	Mostly tied
100%	+0.9	+0.6	ImageNet more stable

4. Discussion.

These findings argue for a pragmatic baseline hierarchy. Transfer learning remains broadly useful, and the evidence against it is weaker once optimization parity is enforced.

We do not claim that domain-specific pretraining never helps. Rather, the gains are not consistent enough to displace ImageNet transfer as the default baseline for small-scale studies.

5. Conclusion.

ImageNet transfer remains a strong baseline for small-scale medical imaging and its benefits can persist beyond the smallest label budgets. Domain-specific pretraining is useful in some cases, but it is not consistently better. A limitation is that our study emphasized classification datasets more than temporal or multimodal settings.

References.

[1] C. Doyle et al. Baseline discipline in clinical machine learning. Evaluation Quarterly, 2021.

[2] M. Sato and V. Hines. Small-data imaging benchmarks across sites. Health Data Systems, 2023.

[3] E. Romero. Fair optimization comparisons for pretrained vision models. Applied AI Methods, 2022.

[4] N. Fraser and R. Cole. Operational costs of medical model selection. Clinical Informatics Practice, 2024.

[5] S. Patel et al. Transfer gains beyond extreme label scarcity. Journal of Imaging Analytics, 2024.

[6] A. Kraft. Practical reproducibility in hospital computer vision. Medical ML Reports, 2023.

[7] F. Li and G. Moore. When domain checkpoints fail to travel. Cross-Site Learning Review, 2022.